



**The engineer makes the decision.**  
**The AI does the detective work.**

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This ran on a real production incident.  
It'll run on the next one too.

**88 min → ~4 min with AI Reckoning**

# Production deployment incident

A live release triggers error logs, buzzing Teams alerts, and customer-impact context.





# The Problem

Too many tools. Too much noise. Not enough time.



## Loki Logs

12

```
20:31:12 INFO deploying checkout-service v4.2
20:31:14 INFO build completed in 42s
20:31:16 ERROR payment API timeout after 2s
20:31:16 ERROR checkout-service 500 response
20:31:17 ERROR correlationId=8f31...a7d2
20:31:18 WARN latency p95 crossed 2s
20:31:19 ERROR database connection failed
20:31:20 INFO retrying payment gateway call
```

## Grafana

Checkout Error Rate (5m) ⓘ

24.8% !



## #prod-alerts

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- AlertManager** 20:31  
checkout-service error rate > 20%  
p95 latency > 2s **URGENT**
- Alex** 20:31  
Customers are reporting failures **WARNING**
- Priya** 20:32  
Payment API timing out **URGENT**
- On-call Bot** 20:32  
PagerDuty incident #4821 triggered **P1**

Type a message...



🕒 INCIDENT TIMER

88:00

TIME ELAPSED

## GitLab

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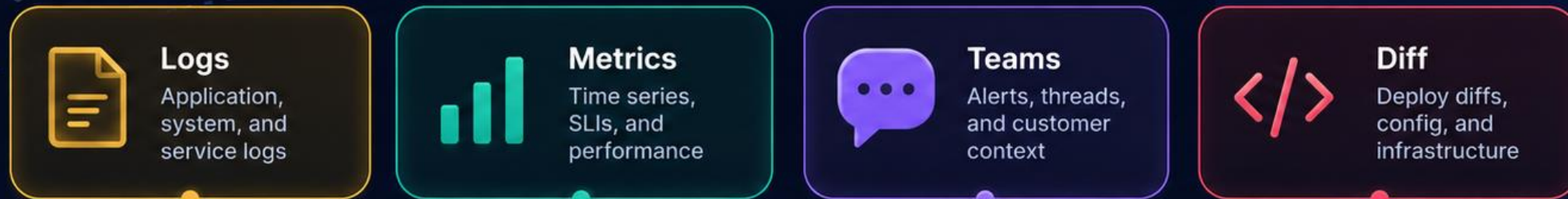
🔗 feat/checkout-timeouts → main **Merged**

checkout-service / src / handler.go

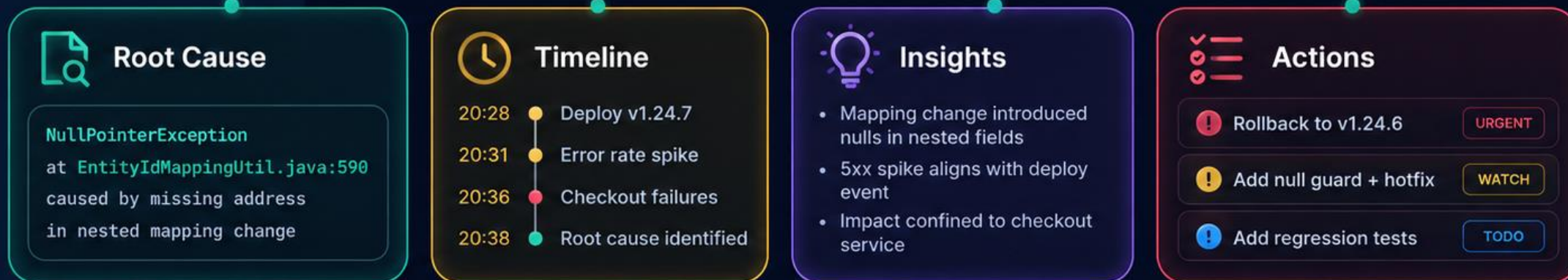
```
@@ -142,7 +142,11 @@ func (h *Handler) Charge() error {
142 - ctx, cancel := context.WithTimeout(ctx,
143 - 2*time.Second)
142 + // Increase timeout to handle gateway slowness
143 + timeout := 2 * time.Second
144 + if featureFlags.IsEnabled("longer-timeout") {
145 +     timeout = 5 * time.Second
146 + }
147 + ctx, cancel := context.WithTimeout(ctx, timeout)
...
```

# What We Built

One button. Four sources. Minutes.

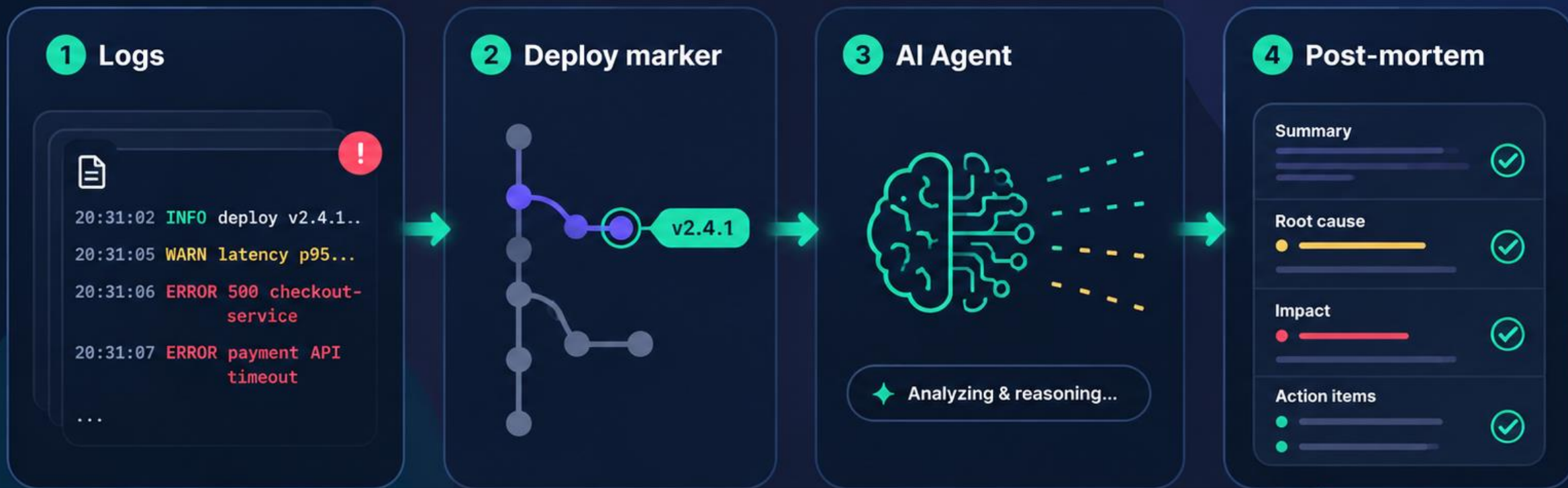


GPT-4o • cross-source correlation





# The Process



# AI-powered incident analysis

An agent turns noisy logs, deploy markers, and customer context into a post-mortem.

● ROOT CAUSE FOUND

## INPUT DATA

682 log lines, deploy marker, alerts, and customer context.

## AI agent

● ● ● live prod logs

```
INFO deploy v1.24.7 completed
WARN retries rising on api-gateway
ERROR 500 checkout-service
ERROR java.lang.NullPointerException
at EntityIdMappingUtil.java:590
TRACE correlationId=8f31...
```

## AGENT REASONING

AI AGENT

- > Ingesting 682 log lines... ✓
- > Correlating deploy event at 20:31... ✓
- > Error spike identified at 20:36 ✓
- > Root cause identified ✓

## ROOT CAUSE

590

structured post-mortem

☐ EntityIdMappingUtil.java:590

HIGH CONFIDENCE

### Causal chain

- 1 Deploy introduced mapping change for nested address fields.
- 2 Null check omitted in toDto() causing NPE on missing address.
- 3 5xx spike triggered bulkheads and visible customer impact.

### Action items

- Rollback to v1.24.6  
Restore service stability **URGENT**
- Add null guard + hotfix  
Validate nested fields **URGENT**
- Add regression tests  
Prevent recurrence **WATCH**

Input data

Deploy marker

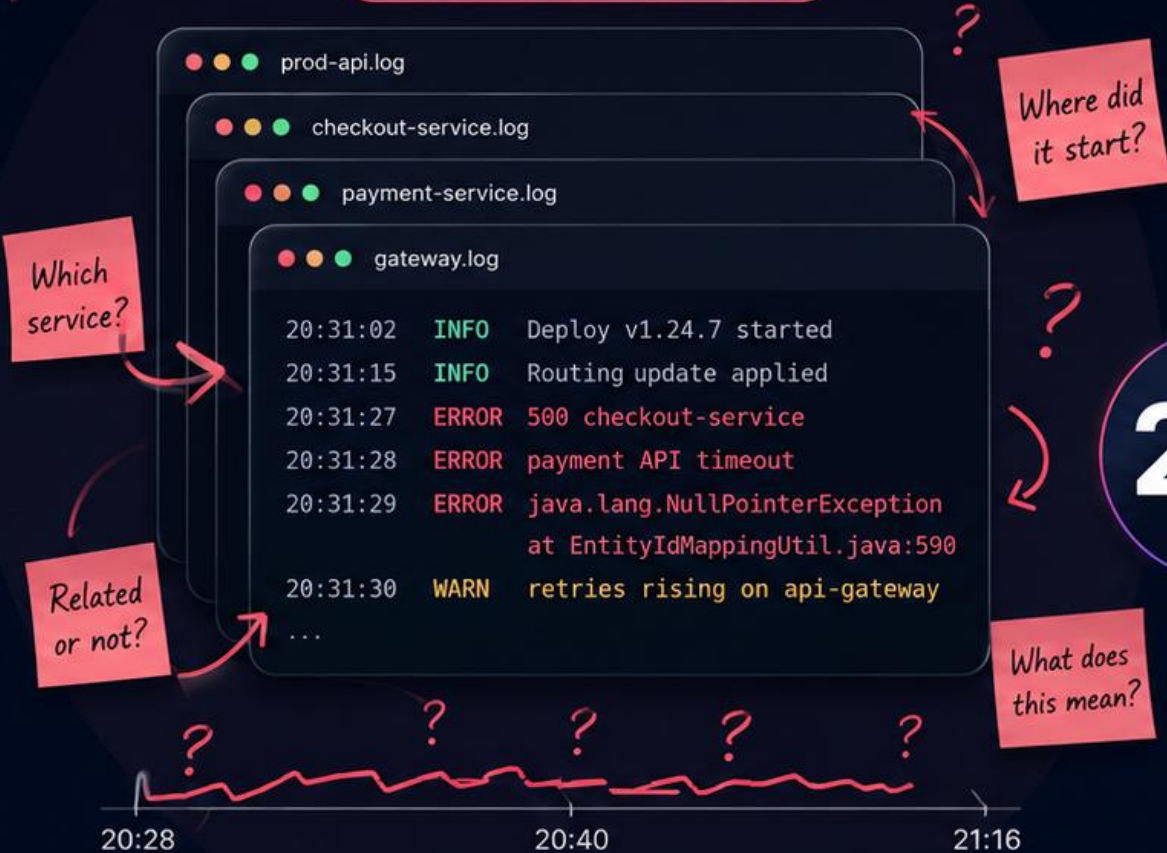
Agent reasoning

Post-mortem

# The Answer: 22x faster from incident to root cause

From manual log hunting to AI-powered root cause analysis and actionable fixes.

🕒 Before — 88 minutes



22x

🕒 After — 4 minutes

## STRUCTURED POST-MORTEM

✓ ROOT CAUSE · HIGH CONFIDENCE

`EntityIdMappingUtil.java:590`

## CAUSAL CHAIN



## IMPACT

- 500 errors for 88 minutes
- Failure rate: 23.6%
- Affected users: 18,231

## ACTION TAKEN

- ✓ Rollback to v1.24.6
- Fix deployed in v1.24.8

88 min

~4 min



# Impact



**88 min**

manual root cause time



**~4 min**

with AI Reckoning



**2.7x**

error rate spike  
while investigating



**14 min**

detection gap —  
deploy to first alert

- ✓ Faster MTTR
- ✓ Instant documentation
- ✓ Faster onboarding





# AI Reckoning

## Incident Investigation, Automated.

From 88 minutes of manual correlation  
to a 4-minute structured report.



Built on a real production incident · Hackathon 2026

live prod logs

```
ERROR 19:21:33 checkout-service 500
ERROR 19:21:33 java.lang.NullPointerException
ERROR 19:21:33 at EntityIdMappingUtil.java:590
WARN 19:21:34 retries rising on api-gateway
ERROR 19:21:35 payment API timeout
ERROR 19:21:35 correlationId=8f3311...
ERROR 19:21:35 java.net.SocketTimeoutException
WARN 19:21:36 latency p95 crossed 2s
ERROR 19:21:37 support thread blocked
ERROR 19:21:37 database connection pool exhausted
ERROR 19:21:38 5xx spike detected
ERROR 19:21:39 null check omitted in toDto()
...
ERROR 19:25:12
ERROR 19:25:13
ERROR 19:25:12
ERROR 19:25:12 cascading failures detected
ERROR 19:25:12 downstream service unavailable
ERROR 19:25:13 user impact increasing
ERROR 19:25:14 manual correlation in progress
...
```

88 min → 4 min



AI-generated post-mortem

STRUCTURED



### Root cause

Null check omitted in toDto() caused NPE on missing address, triggering a cascade of timeouts and 5xx errors.



### Timeline (summary)

- 19:21 Deploy introduced mapping change
- 19:21 Null check omitted in toDto()
- 19:21 Error rate spiked, timeouts started
- 19:22 Retries increased, latency crossed 2s
- 19:23 5xx spike and customer impact
- 19:25 Rollback initiated, service recovery



### Action items

- Rollback to v1.24.6
- Add null guard + hotfix
- Add regression tests
- Improve alert thresholds

URGENT

URGENT

WATCH

DONE